# RUSA Sales & Inventory Analytics

# Bronze → Silver → Gold Data Architecture

Final Architecture Document | Source: uploaded Databricks notebooks

## 1. Architecture Overview

The RUSA Analytics project implements a layered data pipeline in Databricks. Synthetic business data is generated and persisted as raw Delta tables, promoted into a Bronze layer with ingestion metadata, refined into a Silver layer with cleaning and validation, and transformed into Gold analytical datasets used by KPI analysis and dashboard preparation.

| SOURCE / GENERATION | BRONZE | SILVER | GOLD | ANALYTICS / DASHBOARD |
|---|---|---|---|---|
| Python + Pandas<br>NumPy<br>Synthetic data | rusa_analytics_bronze<br>Raw *_raw tables<br>+ _ingestion_timestamp | rusa_analytics_silver<br>Cleaned business entities<br>Validated relationships | rusa_analytics_gold<br>Sales / Inventory /<br>Fulfilment / Risk | KPI Analysis<br>Risk Analysis<br>Databricks Dashboard |

## 2. End-to-End Flow

| Step | Stage | Purpose |
|---|---|---|
| 01 | Data Generation | Creates customers, products, sales orders, inventory, deliveries and billing datasets. |
| 02 | Data Model Validation | Validates relationships, data quality and the Silver model; includes sales-period normalization. |
| 03 | Data Refinement | Builds analytical intermediate datasets such as product demand, inventory risk, slow-moving inventory, delivery a |
| 04 | Gold Layer | Creates persisted Gold analytical datasets for sales, inventory, fulfilment and slow-moving inventory. |
| 05 | KPI Analysis | Calculates sales, inventory, fulfilment and risk KPIs. |
| 06 | Risk Dashboard Preparation | Prepares dashboard-facing tables/views and final risk summary for Databricks visualization. |

# 3. Bronze Layer

**Schema:** rusa_analytics_bronze
The Bronze layer is created from the six raw tables in rusa_analytics. The notebook adds an **_ingestion_timestamp** column to each dataset before writing it as Delta.

| Bronze table | Source raw table | Role |
|---|---|---|
| customers_raw | rusa_analytics.customers_raw | Raw customer records + ingestion timestamp |
| products_raw | rusa_analytics.products_raw | Raw product records + ingestion timestamp |
| sales_orders_raw | rusa_analytics.sales_orders_raw | Raw sales orders + ingestion timestamp |
| inventory_raw | rusa_analytics.inventory_raw | Raw inventory records + ingestion timestamp |
| deliveries_raw | rusa_analytics.deliveries_raw | Raw delivery records + ingestion timestamp |
| billing_raw | rusa_analytics.billing_raw | Raw billing records + ingestion timestamp |

**Bronze principle:** preserve the generated/raw business representation and add ingestion metadata with minimal transformation.

# 4. Silver Layer

**Schema:** rusa_analytics_silver
Silver contains cleaned business entities used by downstream analytical transformations.

| Silver table | Primary role | Key refinement/validation |
|---|---|---|
| customers | Clean customer entity | Deduplicate customer_id; trim text fields |
| products | Clean product master | Deduplicate product_id; trim text; validate positive price/lead time and reorder fields |
| sales_orders | Clean sales transactions | Deduplicate order_id; validate quantity, unit price and sales amount |
| inventory | Clean inventory records | Deduplicate inventory_id; validate stock and reorder levels |
| deliveries | Clean delivery records | Deduplicate delivery_id; require order/date fields; validate delivery timing |
| billing | Clean billing records | Deduplicate invoice_id; require order/date fields; validate invoice amount |

# 5. Gold Layer

**Schema:** rusa_analytics_gold
Gold contains business-facing analytical datasets. The documented principal Gold outputs are sales summary, inventory summary, fulfilment summary and slow-moving inventory. Later notebooks also create dashboard/risk datasets.

| Gold dataset | Purpose |
|---|---|
| sales_summary | Product-level sales aggregation including total quantity sold, total orders, average quantity per order and total_sale |
| inventory_summary | Product inventory position including stock, demand, coverage, inventory status and inventory value. |
| fulfilment_summary | Order/delivery performance and fulfilment metrics. |
| slow_moving_inventory | Identifies products using the project's low-movement threshold and inventory review logic. |
| dashboard_inventory_risk | Dashboard-facing inventory risk information prepared in the final dashboard notebook. |
| dashboard_slow_moving | Dashboard-facing slow-moving inventory information. |
| dashboard_sales | Dashboard-facing sales analysis dataset/view. |

| Gold dataset | Purpose |
|---|---|
| dashboard_inventory | Dashboard-facing inventory analysis dataset/view. |
| dashboard_fulfilment | Dashboard-facing fulfilment analysis dataset/view. |

# 6. Business Logic Flow

| Domain | Inputs | Transformation | Gold output | Consumption |
|---|---|---|---|---|
| Sales | sales_orders + products | Product / order aggregation | sales_summary | Sales KPIs & dashboard |
| Inventory | inventory + products + demand | Coverage, inventory value, status | inventory_summary | Inventory KPIs & risk dashboard |
| Fulfilment | sales_orders + deliveries | Delivery performance & duration | fulfilment_summary | Fulfilment KPIs |
| Slow movement | sales demand + inventory | Average daily demand threshold | slow_moving_inventory | Risk/review dashboard |

# 7. Architecture Notes for Final Handover

• **Source data is synthetic:** the generation notebook creates business datasets using pandas and NumPy with a fixed random seed.

• **Storage format:** the notebooks persist datasets as Delta tables.

• **Bronze:** raw tables are copied into the Bronze schema and enriched with _ingestion_timestamp.

• **Silver:** business entities are deduplicated, trimmed and validated before analytics.

• **Gold:** transformations aggregate the cleaned data into business-facing analytical datasets.

• **Dashboard:** KPI analysis and risk-dashboard preparation consume Gold/refined datasets and create dashboard-facing outputs.

# 8. Validation status / documentation caution

This document describes the architecture represented by the uploaded notebooks. It does not silently mark the pipeline as production-clean. The technical audit identified a fulfilment-date consistency issue, a Gold inventory-risk table-name mismatch, and a syntax issue in one Gold notebook cell. Those are implementation items to resolve before the final project is declared fully reproducible. The architecture itself remains the documented Bronze → Silver → Gold design.

## Source notebooks

• 01_Data_Generation.ipynb

• 02_DataModel_Validation.ipynb

• 03_Data_Refinement.ipynb

• 04_Gold_layer.ipynb

• 05_KPI_Analysis.ipynb

• 06_Risk_Dashboard_Preparation.ipynb